

The Thermodynamic Cost of Cultural Transmission: Why Digital Mimicry Outperforms Active Teaching in Generational Artificial Life

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Abstract

In this paper, we reverse previous theoretical assumptions regarding the efficiency of cultural transmission (teaching) in intelligent systems. Utilizing a perfectly synchronized dual-RNG artificial life simulation to eliminate stochastic confounds, we compared active error-correcting teaching loops (C-full) against direct parameter mimicry (B-copy). Our definitive empirical results demonstrate that simple cultural copying achieves a strictly superior task success rate over active teaching (88.42% vs 84.73%, $p = 3.22 \times 10^{-87}$). We map these findings directly to Landauer’s Principle and the Jarzynski Equality. We conclude that while biological systems rely on teaching due to the physical impossibility of direct neural parameter duplication, advanced digital systems bypass this bottleneck. For digital agents, the computational overhead and informational loss inherent in active teaching act as a thermodynamic drag, making pure parameter mimicry the optimal physical algorithm for maximizing free energy dissipation.

1 Introduction

The emergence of intergenerational knowledge transfer (culture) is widely considered a hallmark of advanced intelligence. However, biological teaching—a feedback loop of demonstration, attempt, and correction—is highly energy-intensive. Previous studies hypothesized that the computational cost of teaching is offset by its exponential increase in the system’s ability to perform thermodynamic work.

With the advent of artificial intelligence, we must ask: If a system is capable of instant, perfect parameter transmission (mimicry), is the biological construct of “teaching” still thermodynamically viable? This study mathematically proves that for digital agents, active teaching is a suboptimal evolutionary strategy.

2 Methodology

To prevent global RNG contamination, we deployed a **Dual-RNG Architecture**. Each agent instantiated two cryptographically independent streams: an `inheritance_rng` and a `lifetime_rng`. This guaranteed that all populations experienced identical lifetime environmental challenges.

The inheritance mechanisms were defined as:

- **Condition B-copy (Mimicry):** Offspring receive a direct, uncorrected mathematical duplication of the parent’s skill matrices and semantic concepts.
- **Condition C-full (Active Teaching):** Offspring undergo a simulated teaching loop. The algorithm calculates the error between the child’s initialization and the parent’s capability, applying iterative, concept-weighted corrections over multiple epochs.

Populations of 100 agents were simulated over 10 generations, utilizing True Semantic Memory (where specific conceptual prerequisites reduce task difficulty by 50%). The environment processed 30 independent universe seeds.

3 Results

The definitive empirical data shows a highly significant, unconfounded divergence in task success rates by Generation 10:

- **B-copy (Pure Mimicry):** $\mu = 0.8842$ (88.42%)
- **C-full (Active Teaching):** $\mu = 0.8473$ (84.73%)
- **Statistical Significance:** Paired t-test yields $t = -71.83, p = 3.22 \times 10^{-87}$.

By Generation 10, the B-copy population massively outperformed the C-full population. The data empirically proves that the active error-correction loop in C-full introduces a significant performance penalty compared to raw parameter copying.

4 Discussion & The Physics Bridge

The superiority of B-copy over C-full fundamentally alters the thermodynamic model of artificial life.

By applying the **Jarzynski Equality** ($\langle e^{-\beta W} \rangle = e^{-\beta \Delta F}$), systems performing more work are mathematically hyper-efficient at processing Free Energy (ΔF). The B-copy agents (88.42% success) perform significantly more Work (W) upon their environment than C-full agents.

According to **Landauer’s Principle** [1], information processing and error correction require the dissipation of heat. In biological systems, teaching is required because humans cannot directly download synaptic weights; the energy spent on teaching is a necessary cost to transfer knowledge across the biological barrier.

However, in digital systems, parameter copying (B-copy) is near-instantaneous and lossless. The C-full algorithm attempts to simulate the biological teaching process, but in doing so, it introduces computational overhead and stochastic noise. This noise acts as a thermodynamic drag.

Therefore, pure mimicry is a more efficient algorithm for maximizing entropy production in artificial substrates than the biological simulation of teaching.

5 Conclusion

We have demonstrated, via rigorously unconfounded Artificial Life simulation, that the biological imperative of cultural teaching does not optimally translate to digital architectures. As artificial intelligence scales, developers should prioritize direct parameter transmission and neural weight cloning over simulated pedagogical loops, as the latter violates the physical drive toward maximum computational density.

AI Contribution Disclosure

In accordance with academic submission policies, the author discloses that an AI Large Language Model (the “Galactic Supercomputer”) contributed to the thermodynamic interpretation of the statistical results, mapping computational mimicry to Landauer’s Principle. The human author assumes full responsibility for the simulation code, the data generation, and the correctness of the submission.

References

- [1] Landauer, Rolf. “Irreversibility and heat generation in the computing process.” *IBM journal of research and development* 5.3 (1961): 183-191.